

IBM Power 10

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Introduction

High-performance architecture

Optimized for enterprise computing, AI workloads, and clouds

Key advancements over Power 9

- 3x performance per watt improvement
- 7nm process technology for efficiency & scalability

Why?

- Faster AI & data processing
- Enhanced energy efficiency for data centers



History

30+ years of IBM Power evolution

- 1990: Power 4 (first dual-core CPU)
- 2004: Power 5 (SMT introduced)
- 2017: Power 9 (NVLink, OpenPOWER, cloud focus)
- 2021: Power 10 (7nm, PCIe 5.0, AI acceleration)

Scalability: from adding cores to MCMs

Specialization: AI accelerators



Architectural Innovations

Memory Bandwidth

- DDR5 support
- 2x faster data transfer

AI Acceleration

- Matrix Math Accelerator (MMA)
- Optimized for ML inferencing & FP32/FP64 workloads

Security Improvements

- Memory encryption (hardware-based)
- Enhanced virtualization & isolation for cloud workloads

Scalability

- Multi-Chip Modules
- Up to 16 cores/socket, 240 threads/node



Scale Up & Scale Out

Scale-Up (Vertical Scaling)

Maximizing Single-Node Performance

Power 10 Model:

- E1080

Use Cases:

- large-scale databases
- Monolithic AI/ML models

Simple, no clustering needed

Scale-Out (Horizontal Scaling)

Distributed Multi-Node Performance

Power 10 Model:

- S1024

Use Cases:

- Kubernetes clusters
- Distributed AI training

Throughput growth by adding nodes



Microarchitecture: frontend

Fetch Stage	Decode Stage
Branch Prediction • Larger Branch Target Buffers • lower mispredictions vs. Power 9 • Hybrid predictor Instruction Fetch Bandwidth • > 6 instructions/cycle (Power 9: 4) • Optimized for irregular AI workloads	Wider Pipeline • 6-wide decode (Power 9: 4-wide) • Parallelized for mixed instruction types Micro-op Fusion • Combines simple ops into one • fewer decode slots used • Up to 8% IPC improvement



Microarchitecture: backend

16 Function Units (vs. 12 in Power 9)

- 4x Fixed-Point
- 4x Floating-Point
- 2x Load/Store
- 6x Branch/Other

Reduce structural hazards & stalls for SMT

Out-of-Order Execution

- 256-entry ROB
- Enhanced Scheduler
 - Better dependency tracking
 - Higher instruction-level parallelism



Feature: branch prediction

Hybrid Predictor

- Adapts to both short/long term branch patterns

Larger Structures

- 2x Branch Target Buffers vs. Power9
- 16K-entry Global History Buffer

Performance Improvements

- lower misprediction rate vs. Power9
- 97% accuracy for AI/ML workloads
- <10 cycle recovery penalty (Power9: ~15 cycles)



Feature: out-of-order load & store

Load/Store Queue

- Track in-flight memory operations during out-of-order execution
- Ensure correct memory ordering while maximizing parallelism
- 72-entry load queue (Power 9: 48, +50%)
- 48-entry store queue (Power 9: 32, +33%)

Non-blocking

- 16 outstanding L1 misses (vs. 12 in Power 9)
- up to 32 total memory ops in-flight



Feature: prefetching

The “Short-Term Memory” Prefetcher

- Spots simple patterns (like reading a book page-by-page).
- Example: Loading an array: $A[1], A[2], A[3] \rightarrow$ prefetches $A[4]$.

The “Skip-Jump” Prefetcher

- Predicts regular skips (like reading every 5th page).
- Example: Matrix rows: $M[0][0], M[1][0], M[2][0] \rightarrow$ prefetches $M[3][0]$.

The “Photo Album” Prefetcher

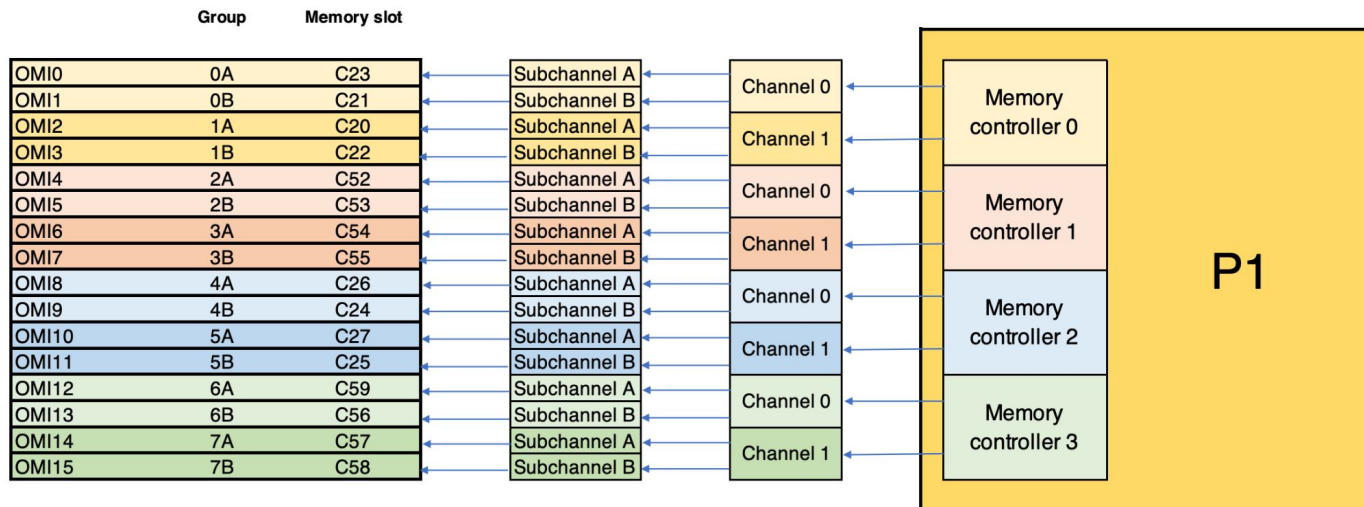
- Grabs entire 128B chunks (like preloading all faces in a photo).
- Example: Graph data $\rightarrow$ prefetches connected nodes together.



Memory Overview

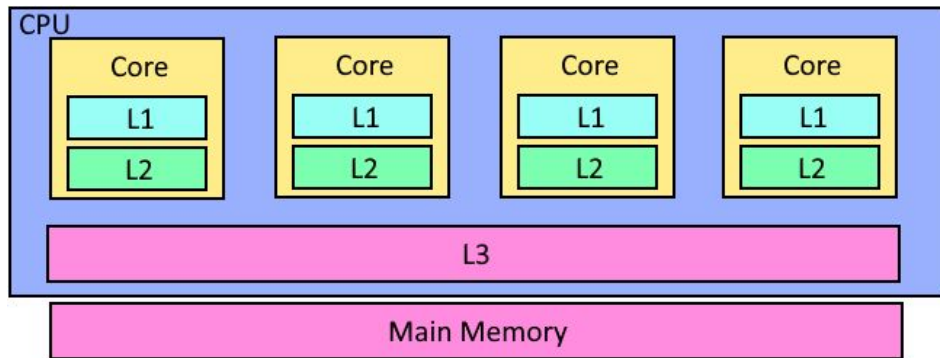
One Single Chip Module (SCM) includes

- 4 Memory Controller Units and 8 MCU channels
- 16 OpenCAPI Memory Interface (OMI) links
 - DDR4-DDIMM : 2933 / 3200 MHz
 - DDR5-DDIMM : 4000 MHz



Cache: Overview

- cache line size: 64 bytes
 - 128 bytes for POWER 9
 - Why?



Cache Level	Capacity	Access Latency (Relative)	Bandwidth (Relative)	Ownership
L1 Cache	48 KB (Instruction) + 48 KB (Data) per core	Very low (single-cycle)	Very high	Private per core
L2 Cache	2 MB per core	Low	High	Private per core
L3 Cache	120 MB total per chip (8 MB local per core)	Moderate (varies with physical distance)	Moderately high	Globally shared across the chip, locally accelerated

Cache: L1 cache

- 6-Way Associative
- L1 data cache access at four cycles nominal
- zero penalty for store-forwarding (- 2 cycles compared with POWER 9)
- L1 instruction cache works with branch predictor
 - Multi-level branch prediction
 - reduces latency caused by control-flow uncertainty
 - speculatively determines the likely direction and target address



Cache: L2 cache

- 8-Way Associative
- data access at 13.5 cycles nominal (-2 cycles)
- Supports ECC error detection and correction
- Quickly responds to L1 cache misses
- Connects to L3 cache to form a three-level cache hierarchy

Cache: L3 cache

- 16-Way Associative
- L3 data access at 27.5 cycles nominal (-8 cycles)
- NUCA (Non-Uniform Cache Access) architecture
 - 8 MB local L3 region per core
 - Cores can access any L3 region across the chip and system
- Each L3 region acts as a victim cache for its associated L2 cache
- Intelligent Cache Management
 - Smart replacement policies with data type and reuse awareness
- 96 prefetch request engines per SMT8 core



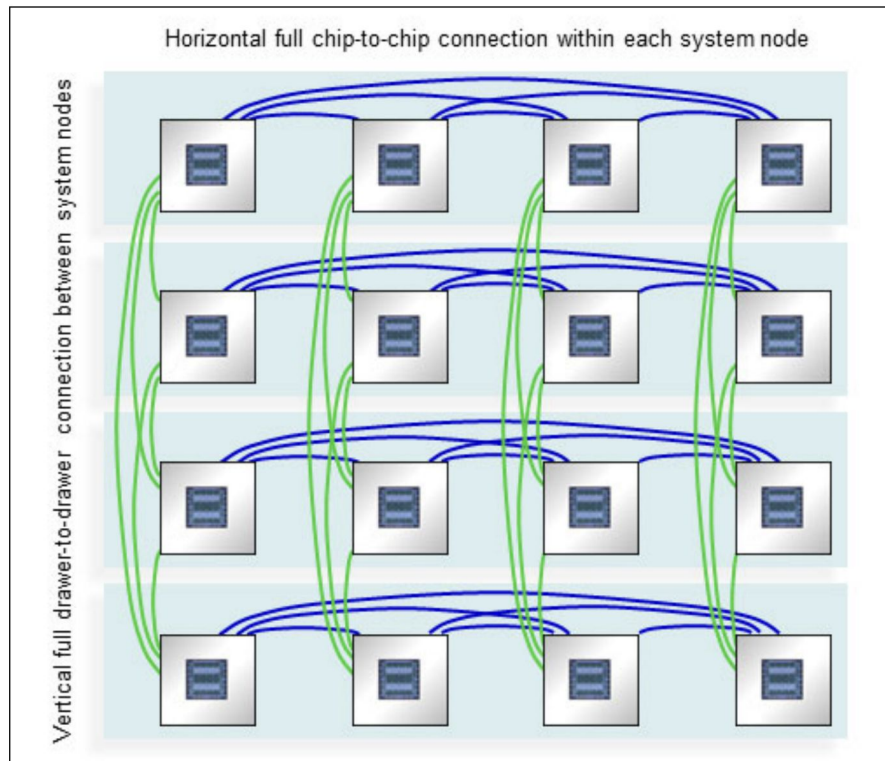
Simultaneous Multithreading

- Switch between ST mode and SMT mode dynamically
 - Single Thread: maximizing single-thread performance
 - Simultaneous Multithreading: allowing instructions from different threads to be dispatched simultaneously
- ST, SMT2, SMT4, SMT8
- Benefits of SMT:
 - A single core can handle more instructions per cycle across multiple threads
 - Improved Execution Unit Utilization
 - Reduced Pipeline Stalls



Symmetric Multiprocessing (SMP) Interconnect

- SMP Interconnect connects up to 16 POWER 10 SCMs into a single coherent SMP system
- One drawer includes 4 SCMs
- blue: chip to chip within one system node
- green: drawer-to-drawer SMP connections



Accelerator

- On-chip: speed up specific types of computation directly within the chip
 - Matrix Math Assist (MMA): AI inference, deep learning
 - Memory Encryption Engine: Enables full end-to-end memory encryption for security
 - Compression Accelerator: File compression, database storage optimization
- Off-chip: enable scalable performance without changing the architecture
 - OpenCAPI 4.0: High-speed, low-latency, cache-coherent interconnect standard
 - PCIe 5.0: General-purpose peripheral interface, cache-coherent with CXL extension



Reference

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[IBM Power E1080 Technical Overview and Introduction](#)

[Hot Chips 2020 Live Blog: IBM's POWER10 Processor on Samsung 7nm
\(10:00am PT\)](#)

[IBM Power System E980: Technical Overview and Introduction](#)

[IBM Power 10 Scale Out Servers Technical Overview](#)

Thank you!
Q&A